

Tuesday, April 28

Exam:

31 57 / 62.5 98
 ←————→
 22

↑ Grades: All B- or above

Hypothetical

(details on Discord and Canvas)

Plan

Proposals due

↳ I'll message w/ feedback

Diffusion today

Muon next

Talks

Ahmet 4-5pm Wednesday @ Davidson

Feyza 4:15-5:15pm Friday @ Estelle 1021

Generative AI

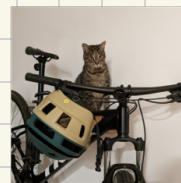
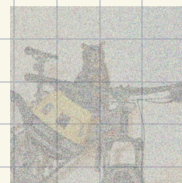
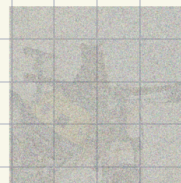
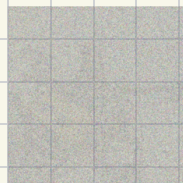
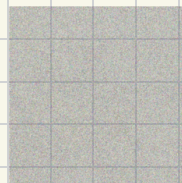
So far: Use VAEs

1. Sample latent from Gaussian
2. Decode

Goal: Scale inference time compute
for better outputs (e.g., reasoning)

Diffusion

$$x_T \sim \mathcal{N}(0, I)$$



training
↙

$$x_0 \sim P_{\text{data}}$$

inference
→

Adding Noise

T steps $t \in \{1, \dots, T\}$ $\beta_t \in (0, 1)$ is noise rate

$$x_t = \sqrt{1-\beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t \quad \text{where} \quad \epsilon_t \sim \mathcal{N}(0, I)$$

$$\begin{aligned} \mathbb{E} \|x_t\|_2^2 &= \mathbb{E} \left[(1-\beta_t) \|x_{t-1}\|_2^2 + \sqrt{1-\beta_t} \sqrt{\beta_t} \langle x_{t-1}, \epsilon_t \rangle + \beta_t \|\epsilon_t\|_2^2 \right] \\ &= (1-\beta_t) \cdot 1 + 0 + \beta_t \cdot \mathbb{E} \|\epsilon_t\|_2^2 = 1 \end{aligned}$$

Fact: $y_1 \sim \mathcal{N}(\mu_1, \sigma_1^2 I)$, $y_2 \sim \mathcal{N}(\mu_2, \sigma_2^2 I)$

$$y_1 + y_2 \sim \mathcal{N}(\mu_1 + \mu_2, (\sigma_1^2 + \sigma_2^2) I)$$

$$x_t \sim \mathcal{N}(\sqrt{1-\beta_t} x_{t-1}, \beta_t I)$$

When T is large x_T is indistinguishable from pure noise

Direct Jump Property

$$\text{let } \alpha_t = 1 - \beta_t.$$

$$X_t = \sqrt{\alpha_t} X_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t$$

$$= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} X_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-1}) + \sqrt{1 - \alpha_t} \epsilon_t$$

$\vdots$

$$= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

where $\epsilon \sim \mathcal{N}(0, I)$

$$X_t \sim \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, \sqrt{1 - \bar{\alpha}_t} I)$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

Removing Noise

Q: What is the distribution to remove noise?

← Must condition on initial image x_0

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t \mathbf{I})$$

$$\text{where } \tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0$$

... But how do we know x_0 ?

Idea: Train model to learn $x_0 \sim p_\theta(x_{t-1} | x_t)$

$$= \mathcal{N}(\mu_\theta(x_t, t), \tilde{\beta}_t \mathbf{I})$$

$$\begin{aligned}
 \text{Loss} &= \mathbb{E}_{t, x_0, \epsilon} \text{KL} (q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t)) \\
 &= \mathbb{E}_{t, x_0, \epsilon} \frac{1}{2\tilde{\beta}_t} \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|_2^2 \quad \left. \begin{array}{l} \text{Instead of predicting} \\ x_0, \text{ let's predict noise.} \end{array} \right\} \\
 &\approx \mathbb{E}_{t, x_0, \epsilon} \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2
 \end{aligned}$$

Inference:

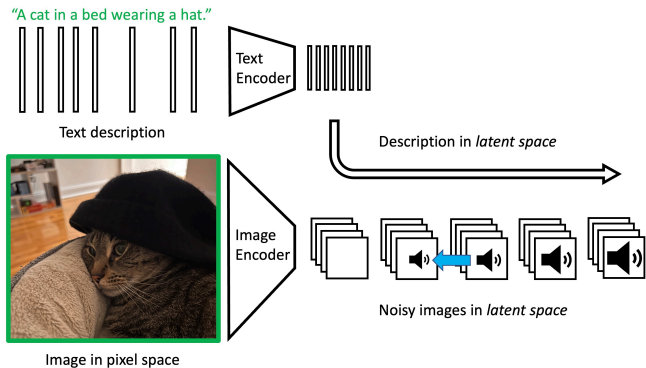
↳ Sample $x_T \sim \mathcal{N}(0, 1)$

↳ For $t = T, \dots, 1$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(x_t, t) \right) + \hat{\beta}_t z$$

↳ Return x_0

Conditional Training

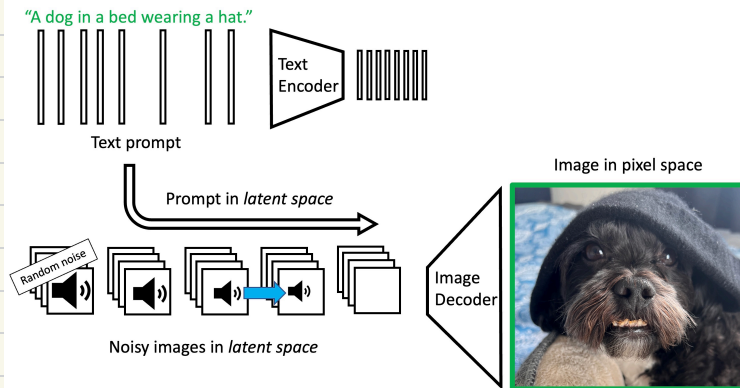


Diffusion model trained to go from more noisy to less noisy encoding *conditioned on description*.

Condition on caption

$$G_{\theta}(x_t, t, c)$$

Conditional Inference



Diffusion model used to go from more noisy to less noisy encoding *conditioned on prompt*.